

MUGHEER ISLAM KHATTAK

Software Engineer | Backend & Data Systems | Analytics Engineering

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PROFESSIONAL SUMMARY

Software Engineering graduate focused on backend systems, data workflows, and production-oriented software development. Hands-on experience building backend applications and APIs using Python, FastAPI, PostgreSQL, SQLAlchemy, and modern engineering practices including Clean Architecture, automated testing, database migrations, CI/CD, and containerized development. Also experienced in analytics engineering and business intelligence, building SQL and dbt-based transformation pipelines, analytical data models, and Power BI dashboards. Seeking opportunities in Software Engineering and Backend Engineering while bringing additional depth in data systems, analytics engineering, and automation.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript, TypeScript

Backend & APIs: FastAPI, REST APIs, SQLAlchemy, Alembic, Pydantic, JWT Authentication, Role-Based Authorization

Databases: PostgreSQL, Relational Data Modeling, Database Migrations, SQL Querying and Optimization

Software Architecture: Clean Architecture, Modular Monolith Design, Repository Pattern, Service Layer, Dependency Injection

Testing & Code Quality: Pytest, Automated Testing, Ruff, pytest-cov, pre-commit

Data & Analytics Engineering: SQL, dbt, dbt Testing, ETL/ELT, Data Modeling, Dimensional Modeling, Data Validation and Quality

Business Intelligence: Power BI, DAX, Power Query, Dashboard Development, KPI Reporting

DevOps & Engineering: Docker, Git, GitHub, GitHub Actions, CI/CD, Linux

Web & Mobile: React, TypeScript, React Native, Expo, Firebase

EXPERIENCE

Independent Software & Data Engineering Practice

Self-Directed Project-Based Work | May 2025 – Present | Remote, Pakistan

Engaged in self-directed software and data engineering work, building projects that apply backend development, data engineering, analytics, testing, and modern software engineering practices.

- Designed and developed backend and application systems using Python, FastAPI, PostgreSQL, React, and TypeScript.
- Applied Clean Architecture principles, structured application layers, automated testing, database migrations, Docker-based development environments, and CI/CD practices.
- Built analytics workflows covering data preparation, transformation, modeling, and reporting using Python, SQL, dbt, PostgreSQL, and Power BI.
- Developed project-based experience across backend engineering, analytics engineering, business intelligence, and mobile application development.

SELECTED ENGINEERING & DATA PROJECTS

Customer360 — Customer Management Platform

July 2026 – Present

Technologies: Python, FastAPI, PostgreSQL, SQLAlchemy, Alembic, Docker, Pytest, Ruff, GitHub Actions, React, TypeScript

- Built a customer management platform using a modular monolith structure and Clean Architecture principles, separating API, application, repository, and persistence responsibilities.
- Implemented JWT-based authentication, password hashing, role-based authorization, protected routes, and authenticated user context.
- Developed customer management functionality including search, filtering, sorting, pagination, customer status management, validation, notes, tasks, and activity timeline features.
- Implemented PostgreSQL persistence using SQLAlchemy and Alembic-managed database migrations, with explicit transaction and rollback handling.
- Established automated testing and engineering quality practices using Pytest, Ruff, pre-commit hooks, dependency auditing, Docker, and GitHub Actions CI.
- Built an initial React and TypeScript frontend with authentication flows, protected routes, centralized API integration, form validation, and feature-oriented organization.

GitHub: github.com/Mugheerik/customer360-platform

SaaS Funnel & Revenue Analytics

April 2026

Technologies: SQL, dbt, Power BI

- Built analytical models for customer lifecycle analysis, conversion funnels, and revenue performance using SQL and dbt.
- Developed transformation workflows that convert raw business data into structured analytical datasets and reporting outputs.
- Created Power BI reporting for customer, funnel, revenue, and business performance analysis.

Healthcare Analytics Dashboard

July 2025

Technologies: Python, PostgreSQL, dbt, Power BI

- Built data transformation workflows converting raw healthcare datasets into structured analytical models and query-ready reporting layers.
- Developed dbt-based transformation pipelines and Power BI dashboards for healthcare, revenue, operational, and performance KPI reporting.

Stock Market Analytics Pipeline

July 2025

Technologies: Python, SQL, PostgreSQL, dbt, Power BI

- Built data workflows for extracting, cleaning, transforming, and analyzing financial datasets using Python and SQL.
- Developed reusable dbt models and Power BI visualizations supporting market trend and performance analysis.

News Broadcast Updates — Final Year Project

May 2024 | Team of 3

Technologies: React Native, Expo, Firebase, Python, Flask

- Developed a cross-platform mobile application for location-based and community-driven news updates.
- Implemented user authentication, news browsing, user-generated news submission, profile management, saved news functionality, and multimedia support.
- Integrated Firebase Authentication, Firestore, and Firebase Storage; contributed to administrative content moderation workflows.
- Contributed to AI-assisted video summarization functionality using Python and Flask-based processing.
- Worked within a three-person software engineering team, contributing to application development and project task coordination.

EDUCATION

Bachelor of Science in Software Engineering

COMSATS University Islamabad, Abbottabad Campus | 2021 – 2025

CGPA: 2.96 / 4.00

CERTIFICATIONS

- Google IT Automation with Python — Google
- IBM DevOps & Software Engineering — IBM
- Google Advanced Data Analytics — Google
- Google Data Analytics — Google
- Google Business Intelligence — Google
- Microsoft Power BI Data Analyst — Microsoft

LANGUAGES

English — Professional Working Proficiency | Urdu — Native | Pashto — Native / Professional Proficiency

PROFESSIONAL DEVELOPMENT

Currently developing knowledge in areas supporting long-term growth in backend and systems engineering, including Cloud-Native Engineering, Platform Engineering, Distributed Systems, Event-Driven Architecture, Infrastructure as Code, Cloud Data Platforms, Data Orchestration, AI-Assisted Software Systems, and System Design.